LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Physical properties, composition and division of atmosphere, International Standard Atmosphere, variation of properties of air with altitude
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Physical properties, composition and division of atmosphere, International Standard Atmosphere, variation of properties of air with altitude.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams on atmospheric layers; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary composition of the Earth's atmosphere?
(b) How does air behave when it flows over a solid body?
(c) Why is the atmosphere divided into distinct layers?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Composition of atmosphere • Physical properties of air • Flow over bodies and boundary layer	L & Ds	PPT, IB	15 min
(b)	• Division of atmosphere • International Standard Atmosphere • Standard sea level values	L & Ds	PPT, PDF	20 min
(c)	• Variation of pressure with altitude • Variation of temperature with altitude • Variation of density with altitude	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of the International Standard Atmosphere in aviation.
(b) Describe how air density changes as altitude increases.
(c) Define the boundary layer in the context of flow over bodies.

12. Home Assignment and Follow Up	Prepare a brief note comparing geometric altitude, absolute altitude, and pressure altitude, referencing the concepts covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Pressure altitude, density altitude, geometric, absolute altitude and standard altitude	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min